

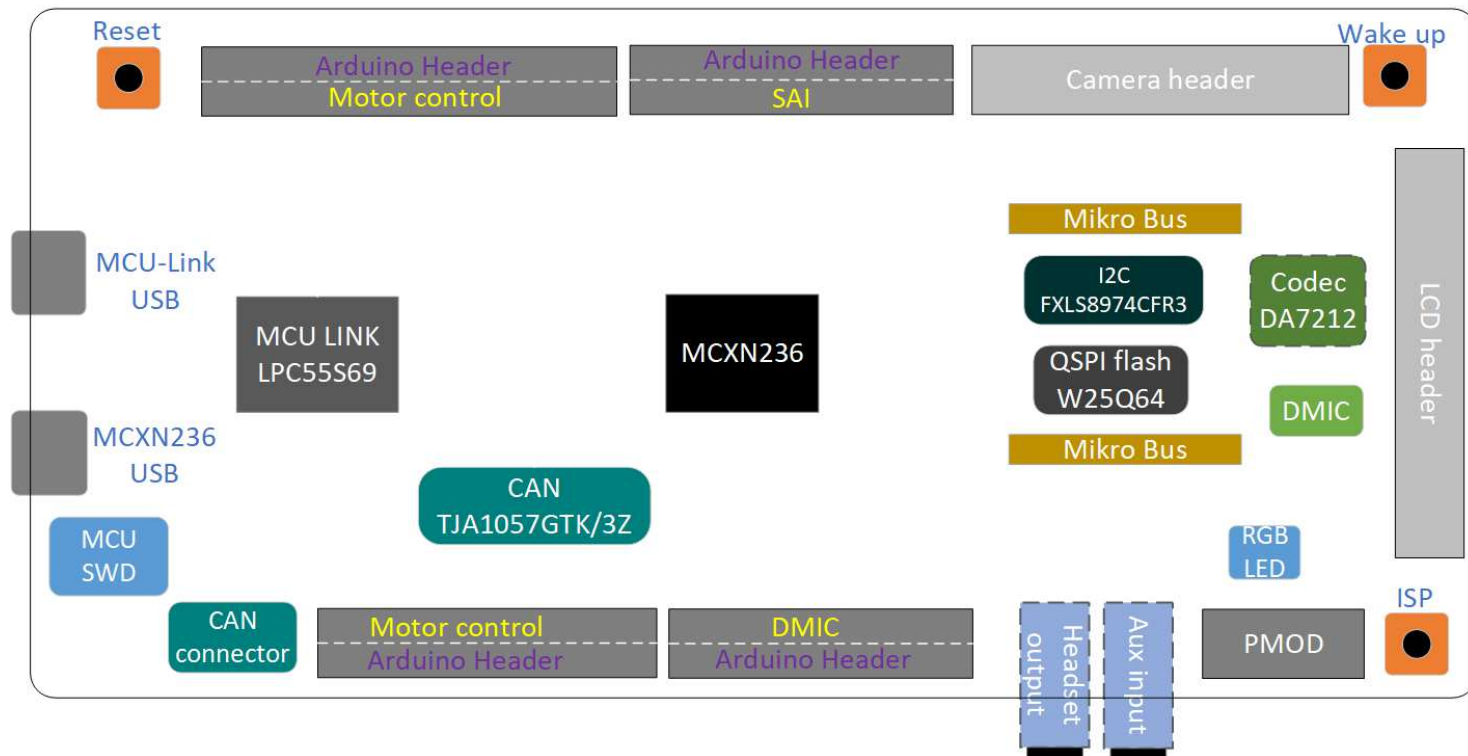
FRDM-MCXN236

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REVISION HISTORY

REVISION	DESCRIPTION OF CHANGE	DATE	Approved
X1	Draft	28-Apr-2023	Jonson Chen
A	Released A085	20-June-2023	Jonson Chen
B	Reverse QSPI D2 and D3 Change R276 from 1M to 0R Add 0R registers R23, R24 for VDD_ANA and VDD_USB wire Update the header index of MikroE, FlexIO, SmartDMA, PMOD interfaces Change the J2-18, J2-20 from P4_0/P4_1 to P1_16/P1_17, add R6 and R7 Change the buttons from 434153017835 to 434123025826 Change ARD D2, D3, D5, D6, D9, D10, D11 from CTIMER to FlexPWM pins Remove DNP setting for resistors of MCU-Link SPI/I2C bridge Remove L5, L16, add R301,C157, R300, C155 Add R8	14-Sep-2023	Jonson Chen
B1	DNP U6, J14, J21. Change C28 from 0.1uF to 2.2uF. Change C18 from 1.0uF to 4.7uF. Change C30 from 0.1uF to 1.0uF	24-Jan-2024	Jonson Chen
B2	Upgrade to Rev B2 due to ARAS system issue, no design change.	26-Jan-2024	Jonson Chen
C	Update signals of J8_6, J8_8. Delete C157, C155, R301, R300 Change R89, R121 from 10K to 0R, DNP R91, R122 Add TP10 - TP32 for automatic testing	25-Mar-2024	Jonson Chen

BLOCK DIAGRAM



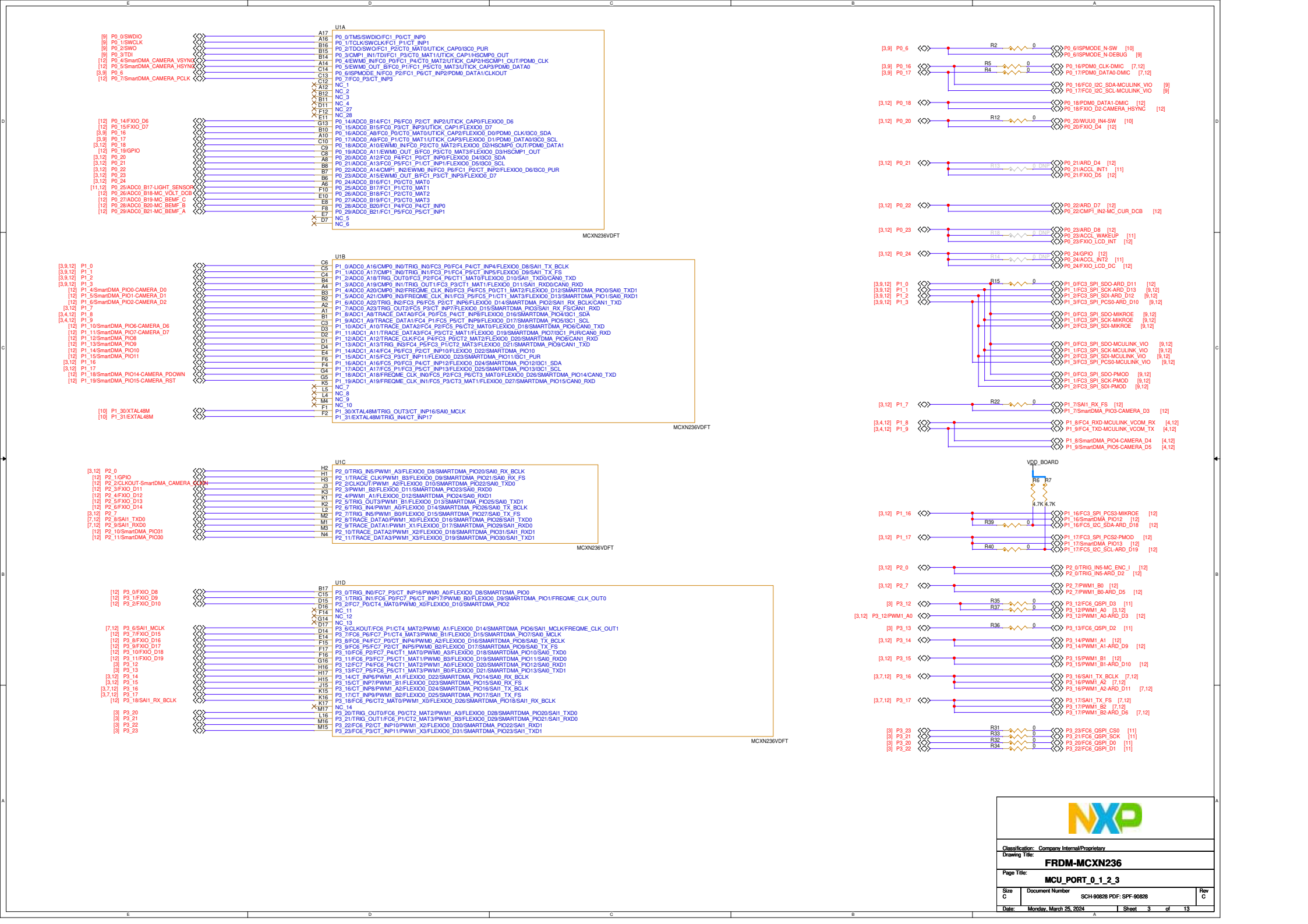
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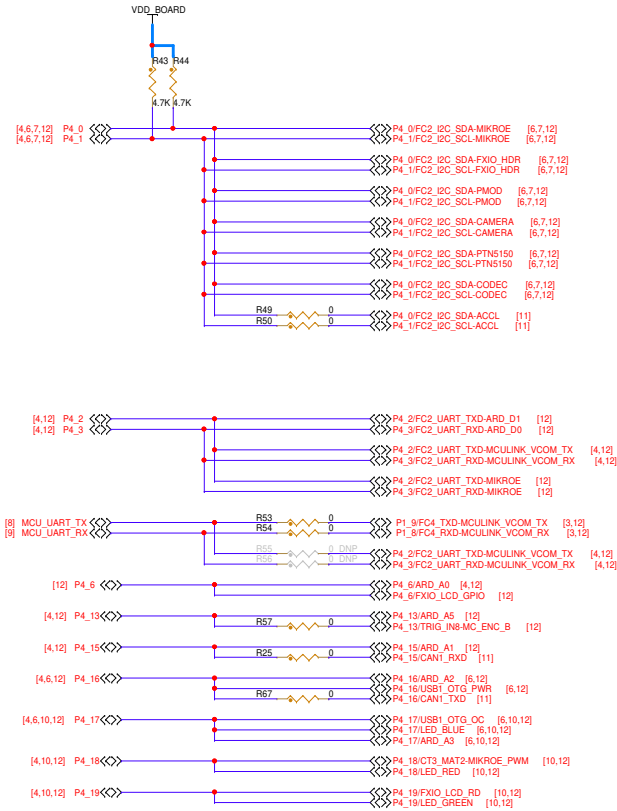
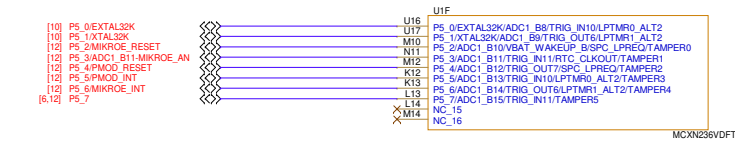
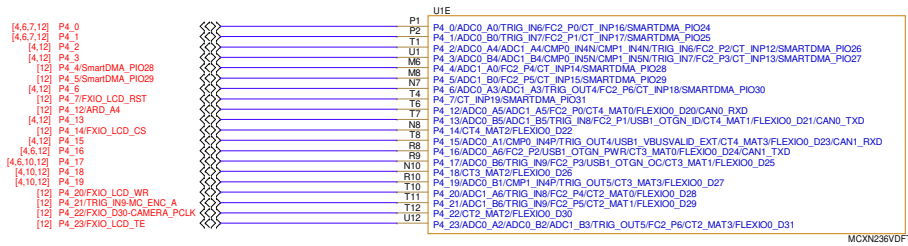
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Page Title: **BLOCK DIAGRAM**

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Drawing Title:
FRDM-MCXN236

Page Title:
MCU_PORT_4_5_A

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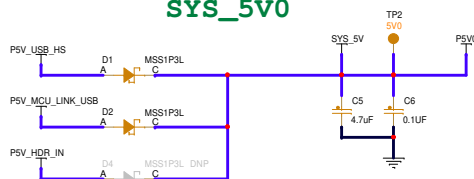
Date: Monday, March 25, 2024

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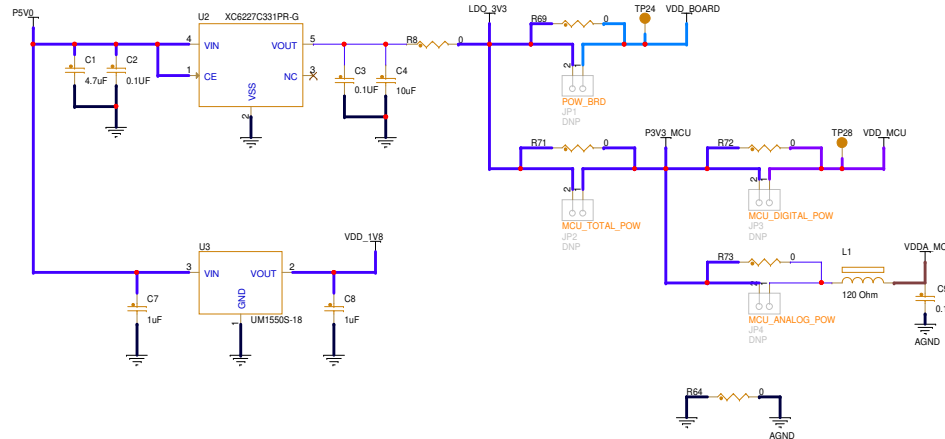
Rev C

POWER

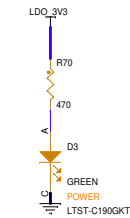
SYS_5V0



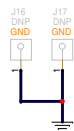
Maximum Output Current : 700mA



POWER LED

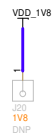


GND Test Points



Note:
Two ground test points are placed
on the left and right sides of the board to
facilitate testing

Power Supply Test Points

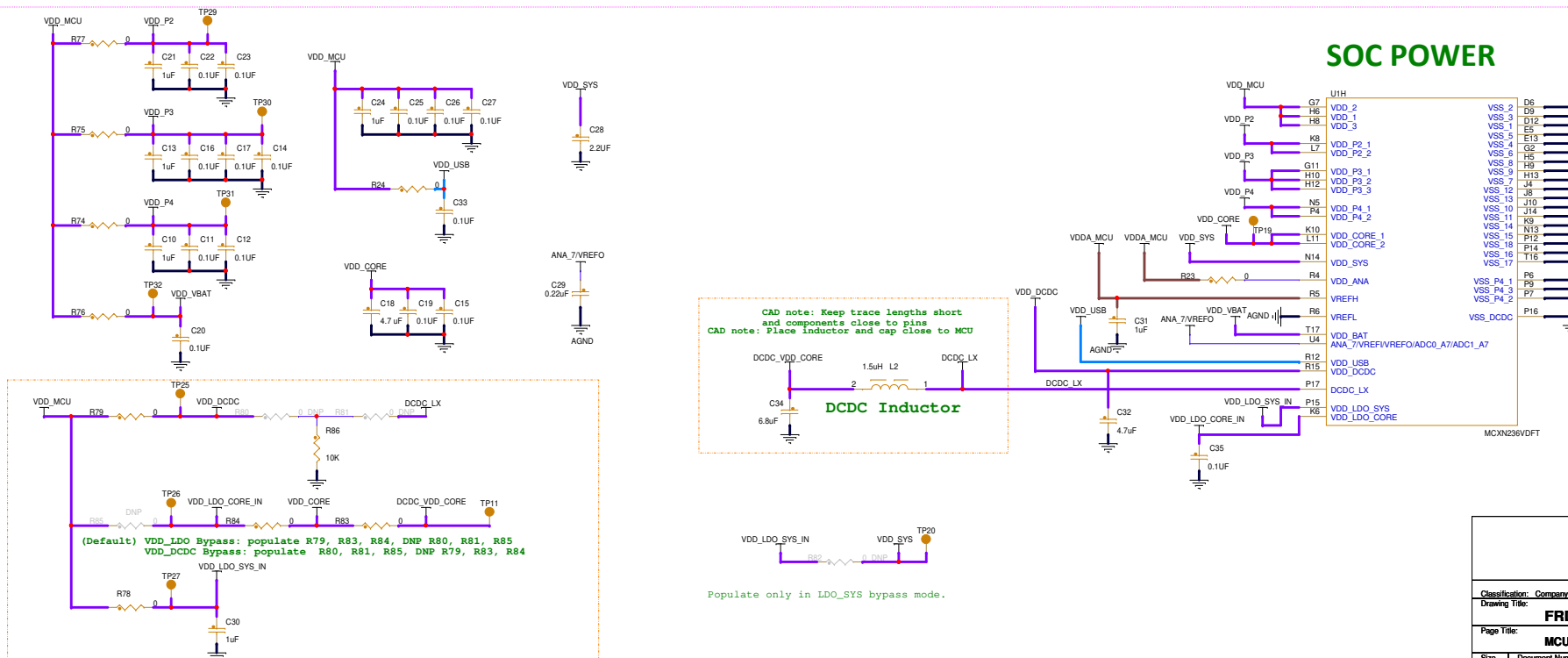


Mounting Holes



Note:
The mechanical mounting holes are
placed on the four vertices of the board

SOC POWER



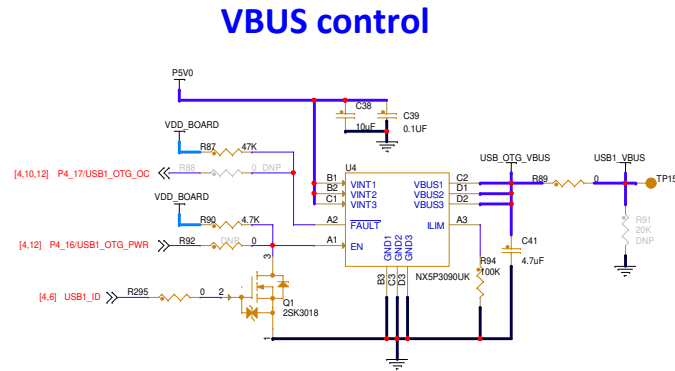
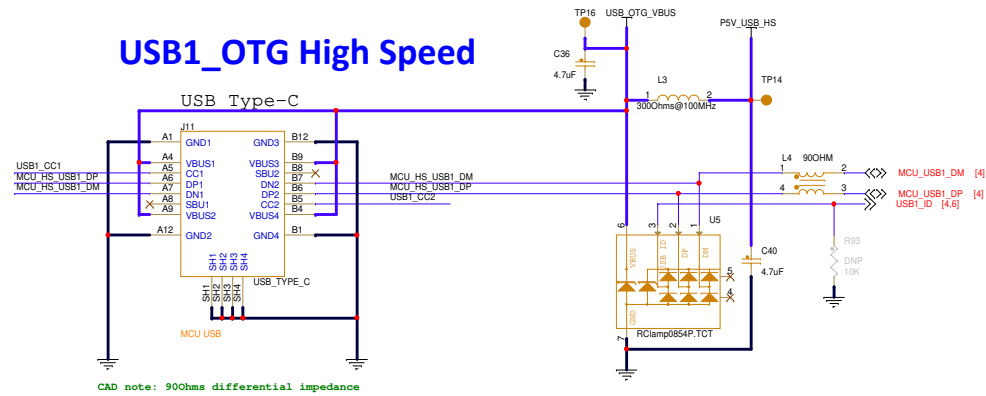
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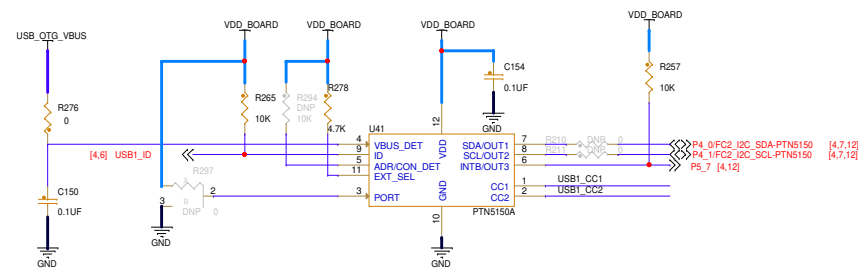
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USB



CC Logic Control



EXT_SEL	PORT	ADR/CON_DET
0---CC2	PORT=1: DFP mode	CON_DET=1, Connection Detected, I2C Address: 0x7A
1---CC1	PORT=0: UFP mode	CON_DET=0, No Connection, I2C Address: 0x3A
	PORT=Float: DRP mode	
Default 1	Float	Float, no I2C function



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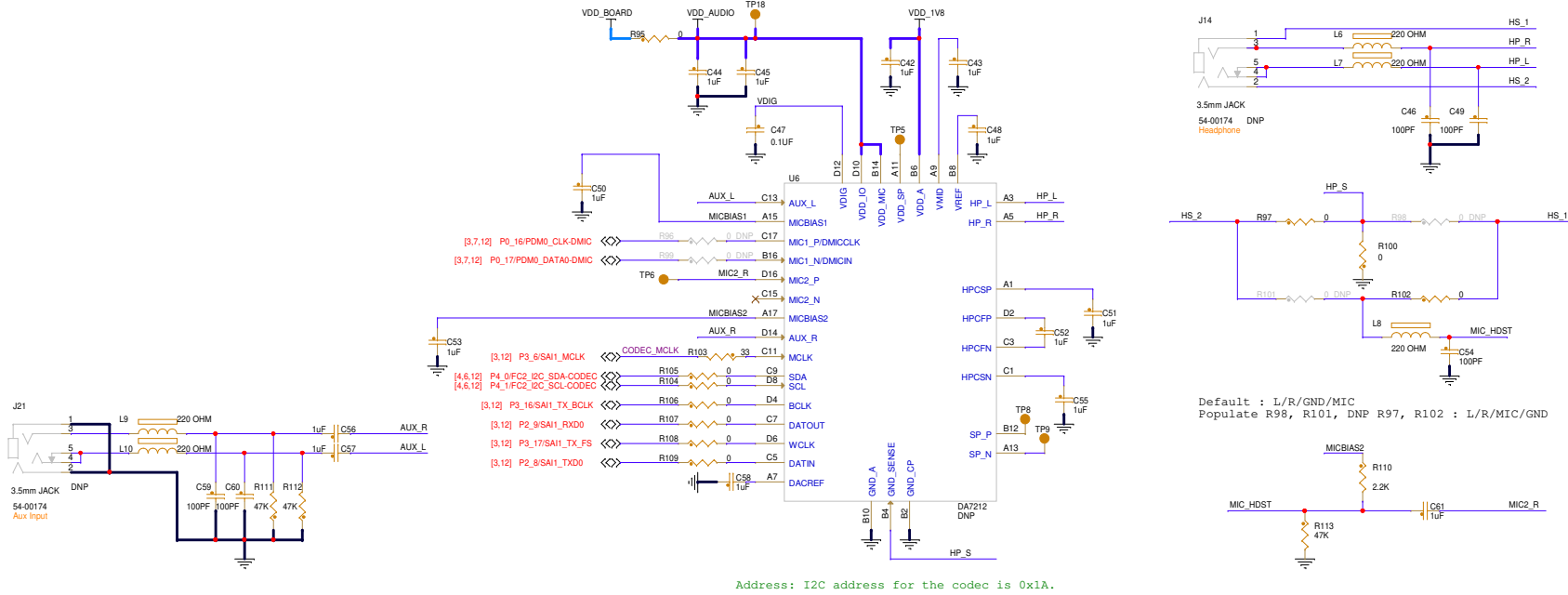
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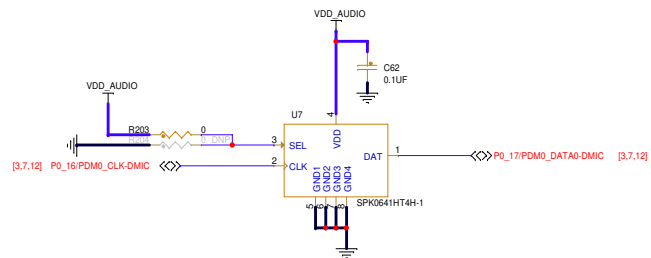
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AUDIO CODEC



DIGITAL MIC



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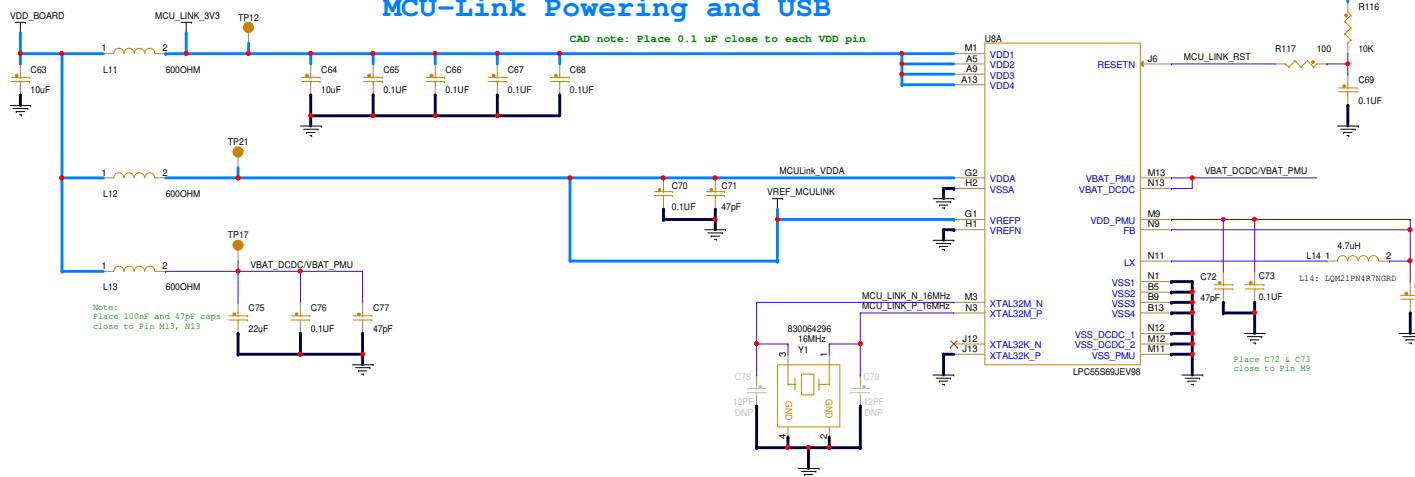
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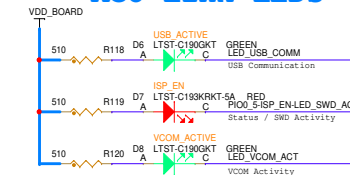
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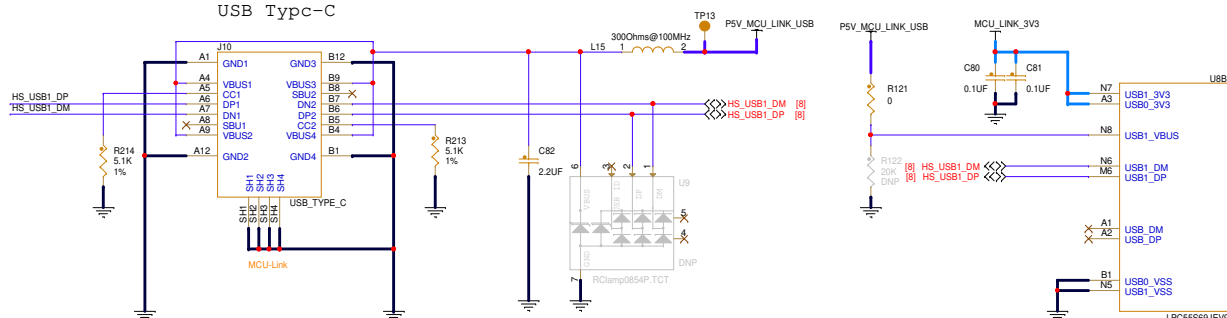
MCU-Link Powering and USB



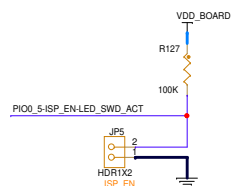
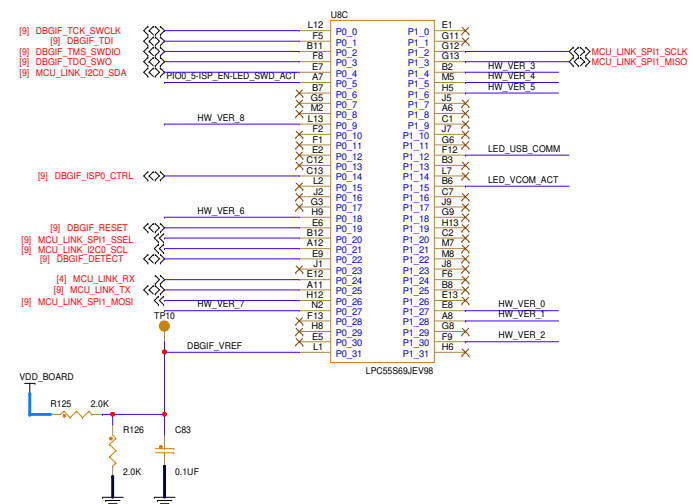
MCU-Link LEDs



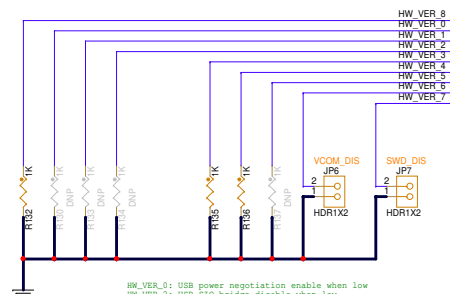
MCU-Link
USB Type-C



CAD note: 900hms differential impedance



Note:
Open : Boot from Internal Flash
Closed: Enable ISP mode (USB1)



```
HW_VER_0: USB power negotiation enable when low
HW_VER_2: USB-SIO bridge disable when low
HW_VER_3: OB selection when low
HW_VER_4: Board identity code used when low
HW_VER_5: Power measurement disable if leave open
HW_VER_6: VCOM disable when low
HW_VER_7: SWD disable when low
HW_VER_8: Secondary VCOM disable when low
```



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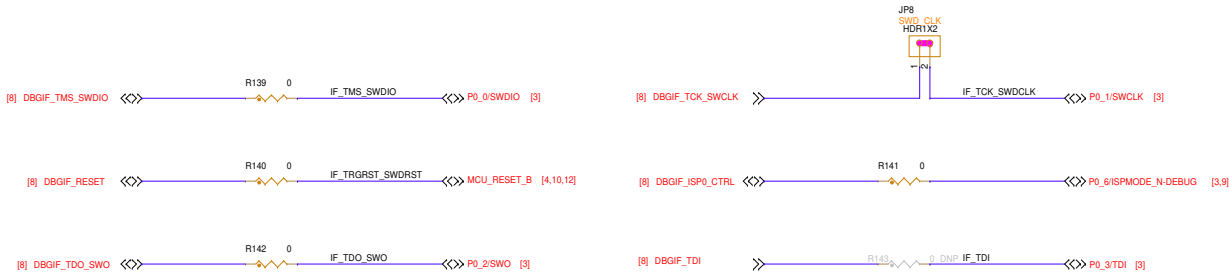
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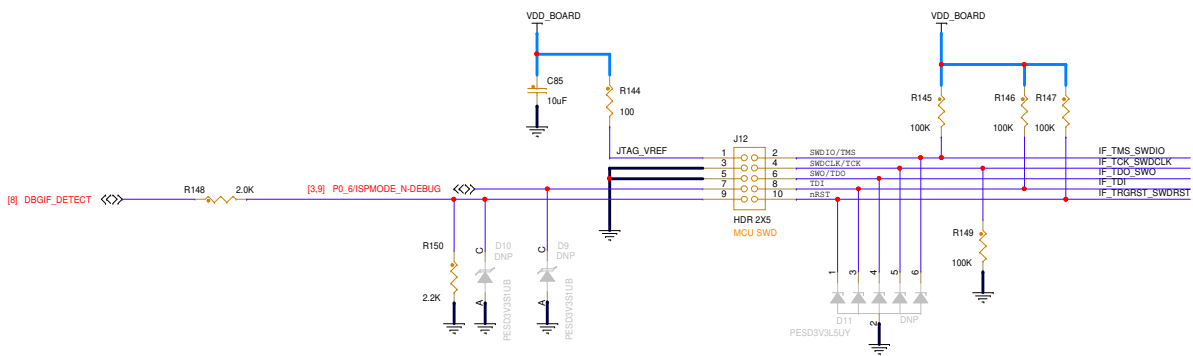
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MCU-Link Debug Interface



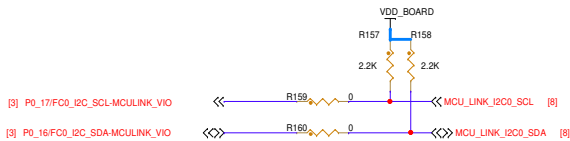
External Debugger



MCU VCOM

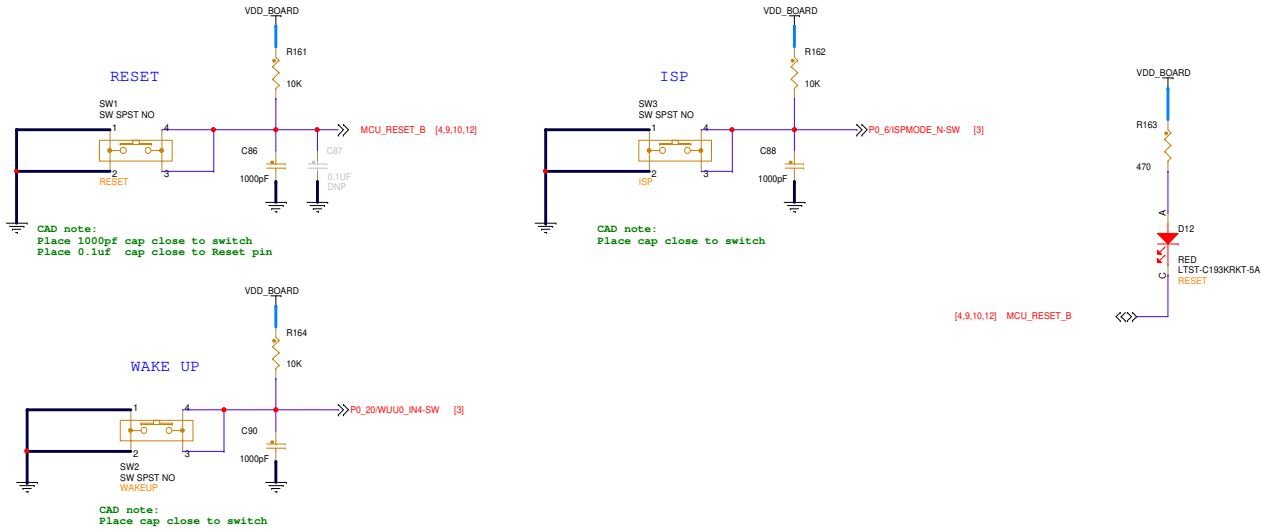


USB Bridge

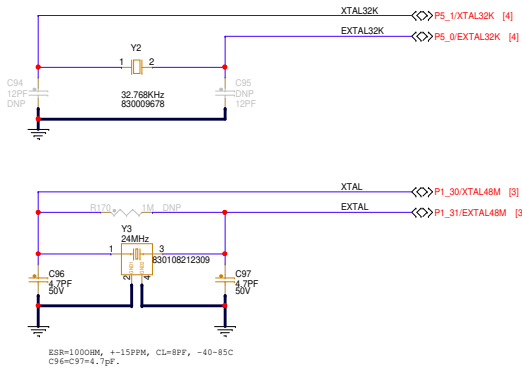


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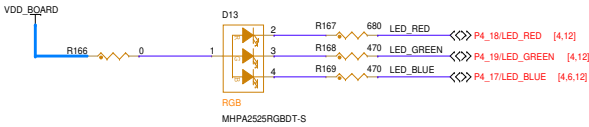
PUSH BUTTON



Clock

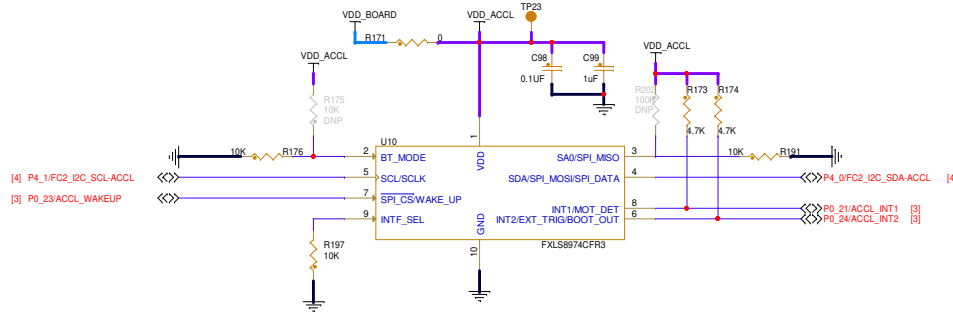


RGB LED



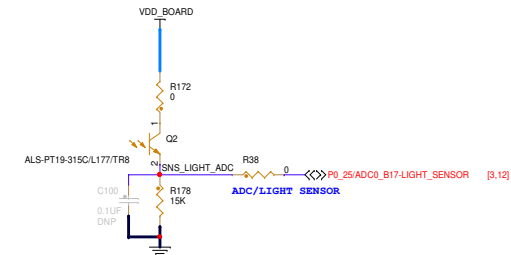
SENSOR & CAN & QSPI

I2C Sensor

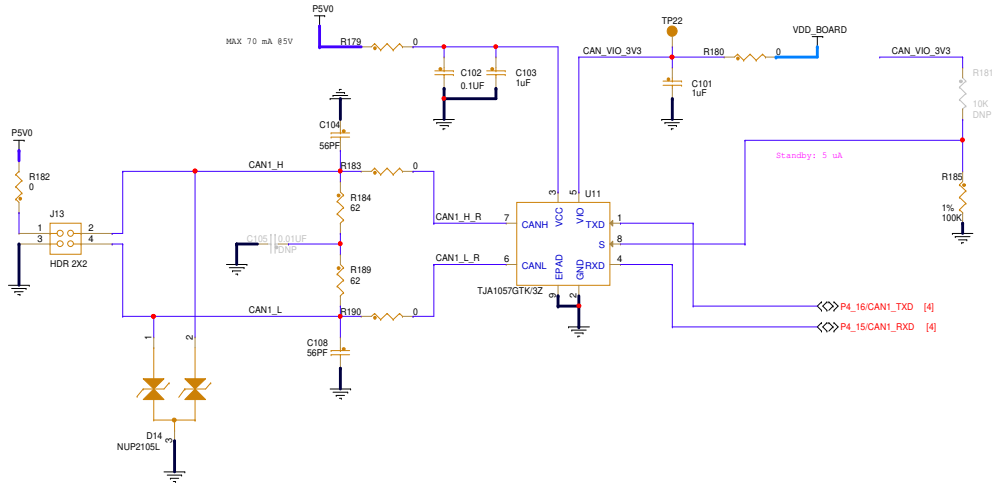


SA0 8-bit I2C write address 8-bit I2C read address
 SA0: 0 30h 31h
 SA0: 1 32h 33h

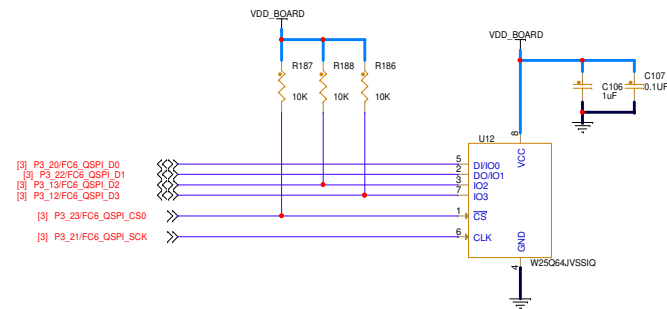
Ambient Light Sensor



FlexCAN



QSPI FLASH



[illegible]

LCD header

Diagram showing the LCD header (J8) connections. The header is a 2x14 pin connector. The left side (pins 1-14) is connected to the board's internal components, and the right side (pins 15-28) is connected to the LCD module. The connections are as follows:

- P1: 1FC2_I2C_SDA-FXIO_HDR [4,6,7]
- P2: 1FC2_I2C_SCL-FXIO_HDR [4,6,7]
- P3: 2FXIO_LCD_INT [3]
- P4: 7FXIO_LCD_RST [4]
- P5: 14FXIO_LCD_CS [4]
- P6: 18FXIO_LCD_RD [4,10]
- P7: 20FXIO_D4 [3]
- P8: 14FXIO_D6 [3]
- P9: 0FXIO_D8 [9]
- P10: 2FXIO_D10 [3]
- P11: 2FXIO_D12 [3]
- P12: 6FXIO_D14 [3]
- P13: 8FXIO_D16 [3]
- P14: 10FXIO_D18 [3]

Camera header

Diagram showing the Camera header (J9) connections. The header is a 2x16 pin connector. The left side (pins 1-16) is connected to the board's internal components, and the right side (pins 17-32) is connected to the camera module. The connections are as follows:

- P1: 14SmartDMA_PIO10 [3]
- P2: 16SmartDMA_PIO12 [3]
- P3: 18SmartDMA_PIO14-CAMERA_P00WH [3]
- P4: 4SmartDMA_PIO4-CAMERA_D0 [3]
- P5: 6SmartDMA_PIO6-CAMERA_D2 [3]
- P6: 8SmartDMA_PIO8-CAMERA_D4 [3,4]
- P7: 10SmartDMA_PIO10-CAMERA_D6 [3]
- P8: 12SmartDMA_PIO12-CAMERA_CLKIN [3]
- P9: 14SmartDMA_PIO14-CAMERA_CLK [3]
- P10: 16SmartDMA_PIO16-CAMERA_RST [3]
- P11: 18SmartDMA_PIO18-CAMERA_D1 [3]
- P12: 20SmartDMA_PIO20-CAMERA_D3 [3]
- P13: 22SmartDMA_PIO22-CAMERA_D5 [3,4]
- P14: 24SmartDMA_PIO24-CAMERA_D7 [3]
- P15: 26SmartDMA_PIO26-CAMERA_CLK [3]
- P16: 28SmartDMA_PIO28-CAMERA_VSYNC [3]
- P17: 30SmartDMA_PIO30-CAMERA_PCLK [3]
- P18: 32SmartDMA_PIO32-CAMERA_D9 [3]

Mikro Bus

Diagram showing the Mikro Bus (J6) connections. The header is a 2x8 pin connector. The left side (pins 1-8) is connected to the board's internal components, and the right side (pins 9-16) is connected to the mikroBUS module. The connections are as follows:

- P1: 18C3_MAT2-MIKROE_PWM [4,10]
- P2: 6MIKROE_INT [4]
- P3: 3FC2_UART_RX0-MIKROE [4]
- P4: 2FC2_UART_TX0-MIKROE [4]
- P5: 1FC2_I2C_SCL-MIKROE [4,6,7]
- P6: 0FC2_I2C_SDA-MIKROE [4,6,7]

PMOD

Diagram showing the PMOD header (J7) connections. The header is a 2x8 pin connector. The left side (pins 1-8) is connected to the board's internal components, and the right side (pins 9-16) is connected to the PMOD module. The connections are as follows:

- P1: 1FC3_SPI_SCK-PMOD [3,9]
- P2: 2FC3_SPI_SDI-PMOD [3,9]
- P3: 0FC3_SPI_SDO-PMOD [3,9]
- P4: 17FC3_SPI_FCS2-PMOD [3]

REF DES	JUMPER(DEFAULT)	PAGE NAME
JP5,JP6,JP7	OPEN	08 MCU_LINK_USB
JP8	1-2	09 MCU_LINK_DEBUG

REF DES	ASSY_OPT	PAGE NAME
R13,R14,R18	DNP	03 MCU_PORT_0_1_2_3
R55,R56	DNP	04 MCU_PORT_4_5_A
D4,J15,J16,J17,J20,JP1,JP2,JP3,JP4,R80,R81,R82,R85	DNP	05 MCU_POWER
R88,R92,R93,R210,R211,R294,R297	DNP	06 USB
J14,J21,R96,R98,R99,R101,R204,U6	DNP	07 AUDIO
C78,C79,R130,R133,R134,R137,U9	DNP	08 MCU_LINK_USB
D9,D10,D11,R143	DNP	09 MCU_LINK_DEBUG
C87,C94,C95,R170	DNP	10 CLOCK_RESET_HMI
C100,C105,R175,R181,R202	DNP	11 SENSOR_CAN_QSPI
C110,C111,C112,C113,J7,J19,R298,R299	DNP	12 HEADERS

REF DES	SHORT(DEFAULT)	PAGE NAME
SJ1,SJ2,SJ3,SJ4,SJ5	1-2	12 HEADERS



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